

Problem

A battery can be rated in ampere-hours (Ah) or watt hours (Wh). The ampere hours can be obtained from the watt hours by dividing watt hours by a nominal voltage of 12 V. If an automobile battery is rated at 20 Ah:

(a) What is the maximum current that can be supplied for 15 minutes?

(b) How many days will it last if it is discharged at a rate of 2 mA?

Step-by-step solution

Step 1 of 2

(a)

An automobile battery is rated at 20 Ah.

Calculate the maximum current that can be supplied for 15 minutes.

$$\begin{aligned} i &= \frac{20 \text{ Ah}}{15 \text{ sec}} \\ &= \frac{20 \text{ Ah}}{\left(\frac{15}{60} \text{ h}\right)} \\ &= 80 \text{ A} \end{aligned}$$

Therefore, the maximum current that can be supplied for 15 minutes is **80 A**.

Step 2 of 2

(b)

The battery discharges at the rate of 2 mA.

Calculate the time taken by the battery to discharge.

$$\begin{aligned} t &= \frac{20 \text{ Ah}}{2 \times 10^{-3}} \\ &= 10,000 \text{ hours} \end{aligned}$$

Calculate the number of days the battery lasts.

$$\begin{aligned} \text{days} &= \frac{10000}{24} \\ &= 416.7 \\ &\cong 417 \text{ days} \end{aligned}$$

Therefore, the number of days the battery will last if it is discharged at the rate of 2 mA is

417 days.